



PROGRAMMING 6212

PROG6212

ABSTRACT
POE PART 1

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ERD Race Day System

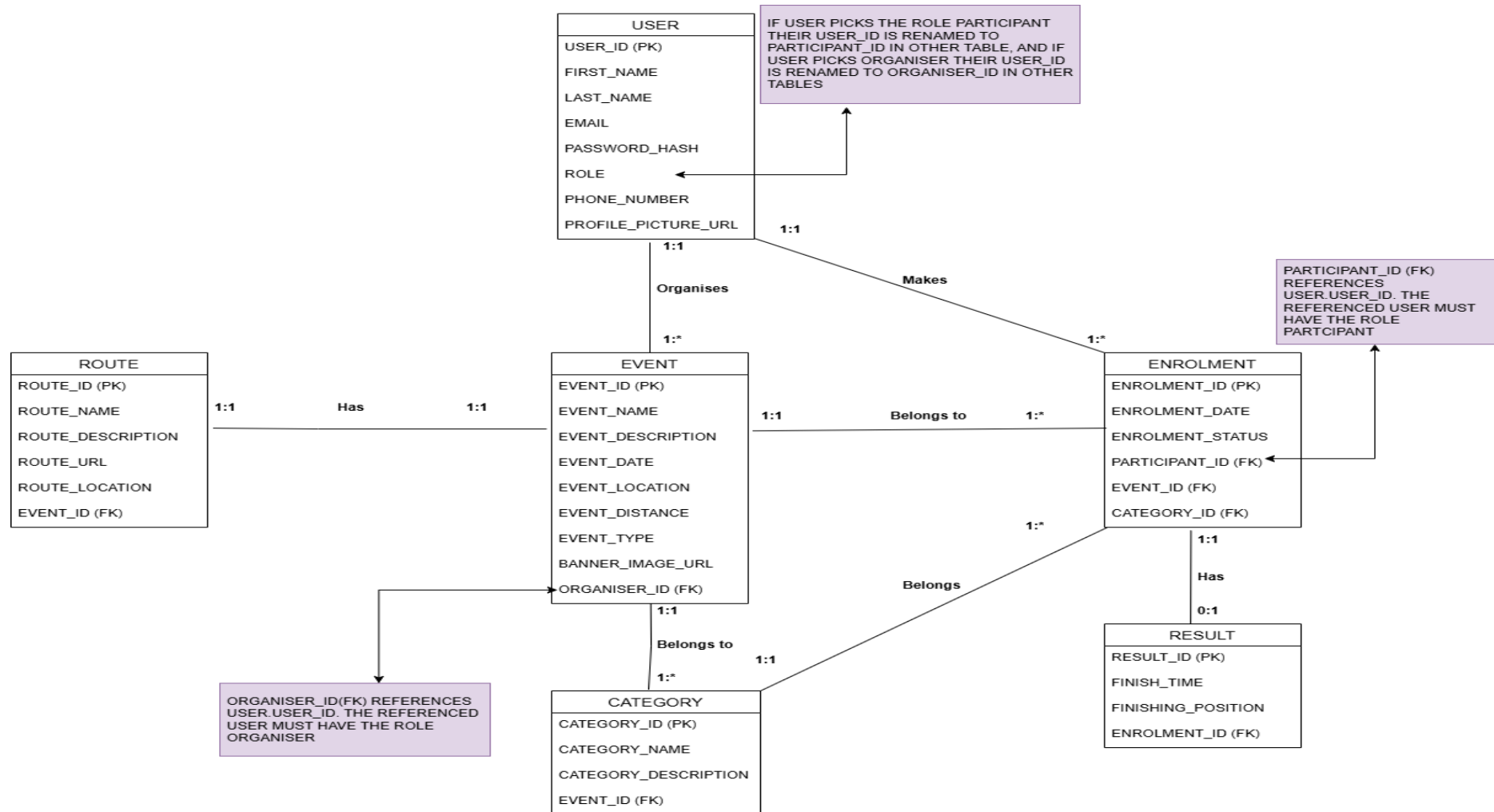


Figure 1 (Coronel, et al., 2020) Entity Relation Diagram

API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Registers a new user as either an Organiser or Participant.	None	FIRST_NAME, LAST_NAME, EMAIL, PASSWORD_HASH, ROLE, PHONE_NUMBER	201 Created – User registered successfully. 400 Bad Request – Invalid details/role. 409 Conflict – Email already exists.
POST	/api/auth/login	Authenticates a user and starts their authenticated session.	None	EMAIL, PASSWORD_HASH	200 OK – Login successful and user/session information returned. 401 Unauthorized – Invalid credentials.
POST	/api/auth/logout	Ends the currently authenticated user's session.	Any	None	200 OK – Logged out successfully. 401 Unauthorized – User is not authenticated.
GET	/api/users/profile	Retrieves the profile of the currently loggedin user.	Any	None	200 OK – User profile returned. 401 Unauthorized – Not logged in.
PUT	/api/users/profile	Updates the currently loggedin user's profile information.	Any	FIRST_NAME, LAST_NAME, EMAIL, PHONE_NUMBER	200 OK – Profile updated. 400 Bad Request – Invalid information. 401 Unauthorized – Not logged in.
GET	/api/events	Retrieves a list of available events.	None	None	200 OK – List of events.

GET	/api/events/{id}	Retrieves the details of a specific event.	None	None	200 OK – Event details. 404 Not Found – Event does not exist.
POST	/api/events	Creates a new event for the loggedin Organiser.	Organiser	EVENT_NAME, EVENT_DESCRIPTION, EVENT_DATE, EVENT_LOCATION, EVENT_DISTANCE, EVENT_TYPE	201 Created – Event created. 400 Bad Request – Invalid data. 401 Unauthorized – Not logged in. 403 Forbidden – User is not an Organiser.
PUT	/api/events/{id}	Updates an event belonging to the loggedin Organiser.	Organiser	EVENT_NAME, EVENT_DESCRIPTION, EVENT_DATE, EVENT_LOCATION, EVENT_DISTANCE, EVENT_TYPE	200 OK – Event updated. 404 Not Found – Event not found. 403 Forbidden – User is not authorised.
DELETE	/api/events/{id}	Deletes an event belonging to the loggedin Organiser.	Organiser	None	204 No Content – Event deleted. 404 Not Found – Event not found. 403 Forbidden – User is not authorised.
GET	/api/events/{eventId} /categories	Retrieves all categories available for an event.	None	None	200 OK – List of categories. 404 Not Found – Event not found.
POST	/api/events/{eventId} /categories	Creates a category for an event managed by the Organiser.	Organiser	CATEGORY_NAME, CATEGORY_DESCRIPTOR	201 Created – Category created. 400 Bad Request – Invalid data. 403 Forbidden – User is not authorised.
PUT	/api/categories/{id}	Updates an existing event category.	Organiser	CATEGORY_NAME, CATEGORY_DESCRIPTOR	200 OK – Category updated. 404 Not Found – Category not found. 403 Forbidden – User is not authorised.

DELETE	/api/categories/{id}	Deletes an event category.	Organiser	None	204 No Content – Category deleted. 404 Not Found – Category not found. 403 Forbidden – User is not authorised.
POST	/api/events/{eventId}/enrolments	Allows a Participant to enrol in an event and select a category.	Participant	CATEGORY_ID	201 Created – Enrolment created. 400 Bad Request – Invalid category. 401 Unauthorized – Not logged in. 403 Forbidden – User is not a Participant. 409 Conflict – Already enrolled.
GET	/api/enrolments/my	Retrieves all events in which the logged-in Participant is enrolled.	Participant	None	200 OK – Participant's enrolments. 401 Unauthorized – Not logged in.
GET	/api/events/{eventId}/enrolments	Retrieves all Participants enrolled in an event managed by the Organiser.	Organiser	None	200 OK – List of enrolments. 403 Forbidden – User is not the event's Organiser. 404 Not Found – Event not found.
GET	/api/enrolments/{id}	Retrieves details of a specific enrolment.	Any authenticated user	None	200 OK – Enrolment details. 403 Forbidden – User cannot access this enrolment. 404 Not Found – Enrolment not found.
POST	/api/events/{eventId}/results	Records a Participant's finishing time and position after an event.	Organiser	PARTICIPANT_ID, FINISH_TIME, FINISHING_POSITION	201 Created – Result recorded. 400 Bad Request – Invalid result. 403 Forbidden – User is not authorised.
GET	/api/results/my	Retrieves the logged-in Participant's	Participant	None	200 OK – Participant's results. 401 Unauthorized – Not logged in.

		personal race results.			
GET	/api/events/{eventId}/results	Retrieves results for Participants in an Organiser's event.	Organiser	None	200 OK – Event results. 403 Forbidden – User is not the event's Organiser. 404 Not Found – Event not found.

Figure 2 (Troelsen & Japikse, 2022) API Endpoint Plan

References

Coronel, C., Morris, S., Crockett, K. & Blewett, C., 2020. *Database Principles: Fundamentals of Design, Implementation, and Management*. 3rd ed. Andover: Cengage Learning.

Troelsen, A. & Japikse, P., 2022. *Pro C# 10 with .Net 6: Foundational Principles and Practices in Programming*. 11th ed. New York: Apress Media LLC.